

first handsfree computing wireless Golden-i headset was demonstrated streaming 480p video at 25fps over Bluetooth 2.0 EDR. It used a speech recognition user interface with approximately 85 per cent efficiency.

Kopin Corporation has over 200 issued patents covering various aspects of Golden-i. Apart from this, the company also has an additional 100+ patent disclosures open, and approximately 60 additional patents filed since 2007, which are being processed.

The team is currently working with the world's leading cellular service providers (such as Verizon Wireless) for bringing the Golden-i Gen 3.8 headsets in various global geographical markets this year.

Design challenges

"The major challenge while developing this product was fitting all the interfaces into a small form factor design," according to Selvaraj Kali-

yappan, general manager - Delivery (PES), Mistral Solutions. The product should offer multiple technologies, such as cameras (infrared, SWIR, near-IR, visible light and terahertz), Wi-Fi, Bluetooth, 4G LTE cellular, 9-axis head tracking and numerous sensors while also ensuring sleek, lightweight and compact design. Space optimisation was achieved by using microBGA and package-on-package (PoP) technology. The Mistral team also addressed multiple routing and layout challenges.

The overall power consumption and heat dissipation in the system was also successfully reduced, thus ensuring longer duration for system operation.

Since Golden-i relies much on voice commands, implementation of a stable voice recognition and active noise cancellation in ambient environments with as much as 120dB of noise was yet another challenge. The algorithm integration was to be performed to

the desired level to meet the specific market demographics and ambient environments that are typically encountered.

Best of its kind

Google Glass is a similar product that made huge headlines last year. But original Golden-i had reached the market ahead of Glass.

"Golden-i is not a single device, but a family of devices leveraging similar hardware and software," says Jeff Jacobsen, senior advisor to CEO, Kopin Corporation, "but significantly differing in headset size and peripherals, depending on the market being addressed."

The initial version of Golden-i 3.8 had been designed for use with hard hats and helmets, for industrial use and applications. Smaller, sleeker and more stylish models of Golden-i are expected to hit the market by 2015. ●

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